Chapter 5. Working with streams

This chapter covers

- Filtering, slicing, and mapping
- Finding, matching, and reducing
- Using numeric streams (primitive stream specializations)
- Creating streams from multiple sources
- Infinite streams

In the previous chapter, you saw that streams let you move from *external iteration* to *internal iteration*. Instead of writing code, as follows, where you explicitly manage the iteration over a collection of data (external iteration),

```
List<Dish> vegetarianDishes = new ArrayList<>();
for(Dish d: menu) {
    if(d.isVegetarian()){
        vegetarianDishes.add(d);
    }
}
```

you can use the Streams API (internal iteration), which supports the `filter` and `collect` operations, to manage the iteration over the collection of data for you. All you need to do is pass the filtering behavior as argument to the `filter` method:

```
import static java.util.stream.Collectors.toList;
List<Dish> vegetarianDishes =
    menu.stream()
        .filter(Dish::isVegetarian)
        .collect(toList());
```

This different way of working with data is useful because you let the ocess the data. As a consequence, the ral optimizations behind the scenes. In ad- the Streams API can decide to run your code in parallel. Using external iteration, this isn't possible because you're committed to a single-threaded step-by-step sequential iteration.

In this chapter, you'll have an extensive look at the various operations supported by the Streams API. You will learn about operations available in Java 8 and also new additions in Java 9. These operations will let you express complex data-processing queries such as filtering, slicing, mapping, finding, matching, and reducing. Next, we'll explore special cases of streams: numeric streams, streams built from multiple sources such as files and arrays, and finally infinite streams.

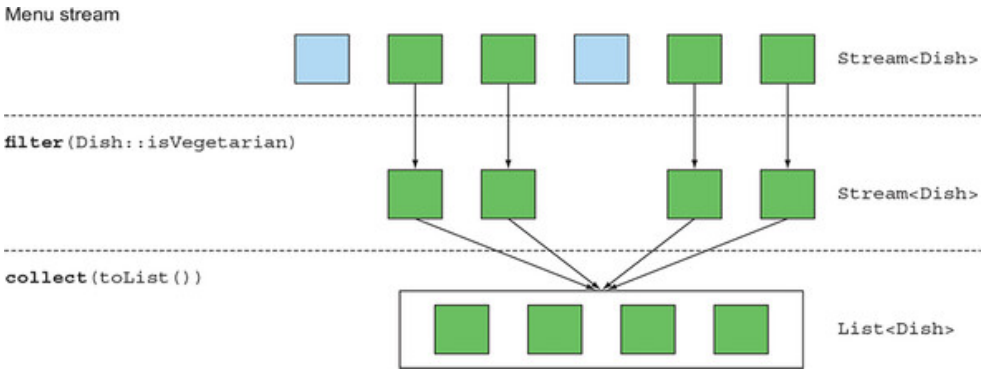
5.1. Filtering

In this section, we'll look at the ways to select elements of a stream: filtering with a predicate and filtering only unique elements.

5.1.1. Filtering with a predicate

The `Stream` interface supports a `filter` method (which you should be familiar with by now). This operation takes as argument a *predicate* (a function returning a `boolean`) and returns a stream including all elements that match the predicate. For example, you can create a vegetarian menu by filtering all vegetarian dishes as illustrated in figure 5.1 and the code that follows it:

Figure 5.1. Filtering a stream with a predicate




```
List<Dish> vegetarianMenu = menu.stream()
                                .filter(Dish::isVegetarian)      1
                                .collect(toList());
```

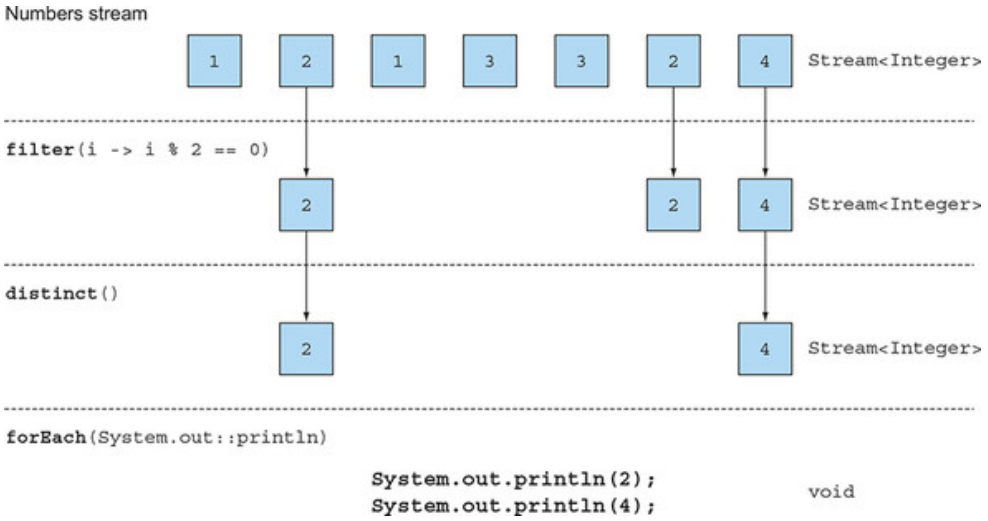
- *1* **Use a method reference to check if a dish is vegetarian friendly.**

nts

l called `distinct` that returns a stream
ng to the implementation of the `hashcode` and `equals` methods of the objects produced by the stream). For example, the following code filters all even numbers from a list and then eliminates duplicates (using the `equals` method for the comparison). Figure 5.2 shows this visually.

```
List<Integer> numbers = Arrays.asList(1, 2, 1, 3, 3, 2, 4);
numbers.stream()
       .filter(i -> i % 2 == 0)
       .distinct()
       .forEach(System.out::println);
```

Figure 5.2. Filtering unique elements in a stream



5.2. Slicing a stream

In this section, we'll discuss how to select and skip elements in a stream in different ways. There are operations available that let you efficiently select or drop elements using a predicate, ignore the first few elements of a stream, or truncate a stream to a given size.

5.2.1. Slicing using a predicate

Java 9 added two new methods that are useful for efficiently selecting elements in a stream: `takeWhile` and `dropWhile`.

Using takeWhile

g special list of dishes:

```
List<Dish> specialMenu = Arrays.asList(
    new Dish("seasonal fruit", true, 120, Dish.Type.OTHER),
    new Dish("prawns", false, 300, Dish.Type.FISH),
    new Dish("rice", true, 350, Dish.Type.OTHER),
```

```
        new Dish("chicken", false, 400, Dish.Type.MEAT),
        new Dish("french fries", true, 530, Dish.Type.OTHER));
```

How would you select the dishes that have fewer than 320 calories? Instinctively, you know already from the previous section that the operation `filter` can be used as follows:

```
List<Dish> filteredMenu
    = specialMenu.stream()
                 .filter(dish -> dish.getCalories() < 320)
                 .collect(toList());                          1
```

- ***1*** **Lists seasonal fruit, prawns**

But, you'll notice that the initial list was already sorted on the number of calories! The downside of using the `filter` operation here is that you need to iterate through the whole stream and the predicate is applied to each element. Instead, you could stop once you found a dish that is greater than (or equal to) 320 calories. With a small list this may not seem like a huge benefit, but it can become useful if you work with potentially large stream of elements. But how do you specify this? The `takeWhile` operation is here to rescue you! It lets you slice any stream (even an infinite stream as you will learn later) using a predicate. But thankfully, it stops once it has found an element that fails to match. Here's how you can use it:

```
List<Dish> slicedMenu1
    = specialMenu.stream()
                 .takeWhile(dish -> dish.getCalories() < 320)
                 .collect(toList());                          1
```

- ***1*** **Lists seasonal fruit, prawns**

Using dropWhile

How about getting the other elements though? How about finding the elements that have greater than 320 calories? You can use the `dropWhile` operation for this:


```
                 .eam()
                 .dropWhile(dish -> dish.getCalories() < 320)
                 .collect(toList());                          1
```

- ***1*** **Lists rice, chicken, french fries**

The `dropWhile` operation is the complement of `takeWhile`. It throws away the elements at the start where the predicate is false. Once the predicate evaluates to true it stops and returns all the remaining elements, and it even works if there are an infinite number of remaining elements!

5.2.2. Truncating a stream

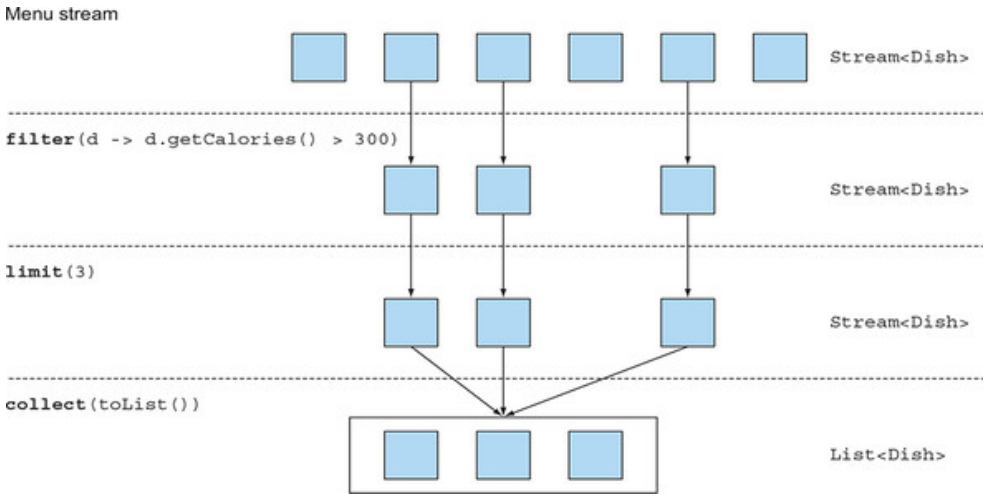
Streams support the `limit(n)` method, which returns another stream that's no longer than a given size. The requested size is passed as argument to `limit`. If the stream is ordered, the first elements are returned up to a maximum of `n`. For example, you can create a `List` by selecting the first three dishes that have more than 300 calories as follows:

```
List<Dish> dishes = specialMenu
                            .stream()
                            .filter(dish -> dish.getCalories() > 300)
                            .limit(3)
                            .collect(toList());                    1
```

- ***1*** **Lists rice, chicken, french fries**

Figure 5.3 illustrates a combination of `filter` and `limit`. You can see that only the first three elements that match the predicate are selected, and the result is immediately returned.

Figure 5.3. Truncating a stream



Note that `limit` also works on unordered streams (for example, if the ...u shouldn't assume any order on the result

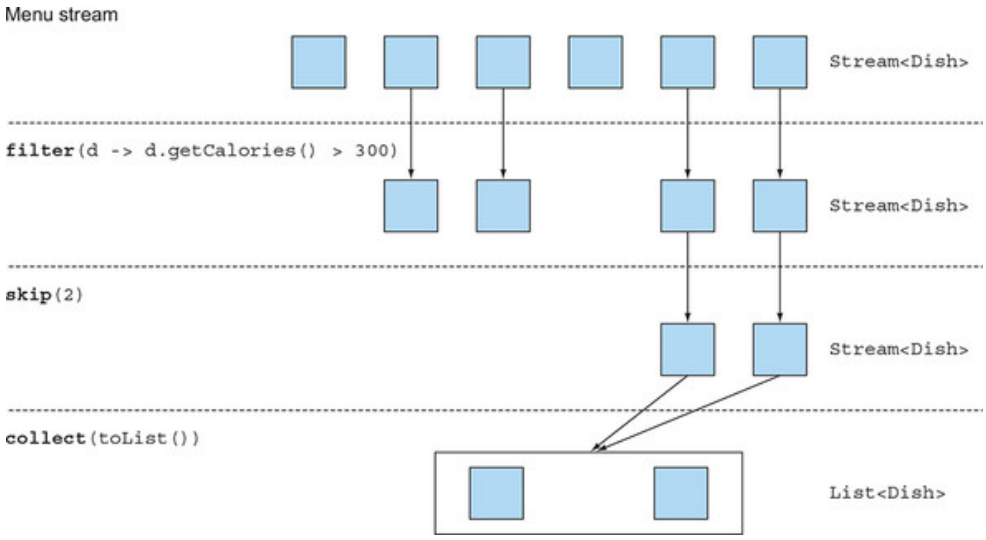
5.2.3. Skipping elements

Streams support the `skip(n)` method to return a stream that discards the first `n` elements. If the stream has fewer than `n` elements, an empty

stream is returned. Note that `limit(n)` and `skip(n)` are complementary! For example, the following code skips the first two dishes that have more than 300 calories and returns the rest. **Figure 5.4** illustrates this query.

```
List<Dish> dishes = menu.stream()
                        .filter(d -> d.getCalories() > 300)
                        .skip(2)
                        .collect(toList());
```

Figure 5.4. Skipping elements in a stream



Put what you've learned in this section into practice with quiz 5.1 before we move to mapping operations.

Quiz 5.1: Filtering
How would you use streams to filter the first two meat dishes?

Answer:

You can solve this problem by composing the methods `filter` and `limit` together and using `collect(toList())` to convert the stream into a list as follows:

```
List<Dish> dishes =
                    -> dish.getType() == Dish.Type.MEAT)
                    .collect(toList());
```

5.3. Mapping

A common data processing idiom is to select information from certain objects. For example, in SQL you can select a particular column from a table. The Streams API provides similar facilities through the `map` and `flatMap` methods.

5.3.1. Applying a function to each element of a stream

Streams support the `map` method, which takes a function as argument. The function is applied to each element, mapping it into a new element (the word *mapping* is used because it has a meaning similar to *transforming* but with the nuance of "creating a new version of" rather than "modifying"). For example, in the following code you pass a method reference `Dish::getName` to the `map` method to *extract* the names of the dishes in the stream:

```
List<String> dishNames = menu.stream()
                             .map(Dish::getName)
                             .collect(toList());
```

Because the method `getName` returns a string, the stream outputted by the `map` method is of type `Stream<String>`.

Let's take a slightly different example to solidify your understanding of `map`. Given a list of words, you'd like to return a list of the number of characters for each word. How would you do it? You'd need to apply a function to each element of the list. This sounds like a job for the `map` method! The function to apply should take a word and return its length. You can solve this problem, as follows, by passing a method reference `String::length` to `map`:

```
List<String> words = Arrays.asList("Modern", "Java", "In", "Action");
List<Integer> wordLengths = words.stream()
                                 .map(String::length)
                                 .collect(toList());
```

Let's return to the example where you extracted the name of each dish. ... the length of the name of each dish? You ...ner `map` as follows:


```
List<Integer> dishNameLengths = menu.stream()
                                    .map(Dish::getName)
                                    .map(String::length)
                                    .collect(toList());
```

5.3.2. Flattening streams

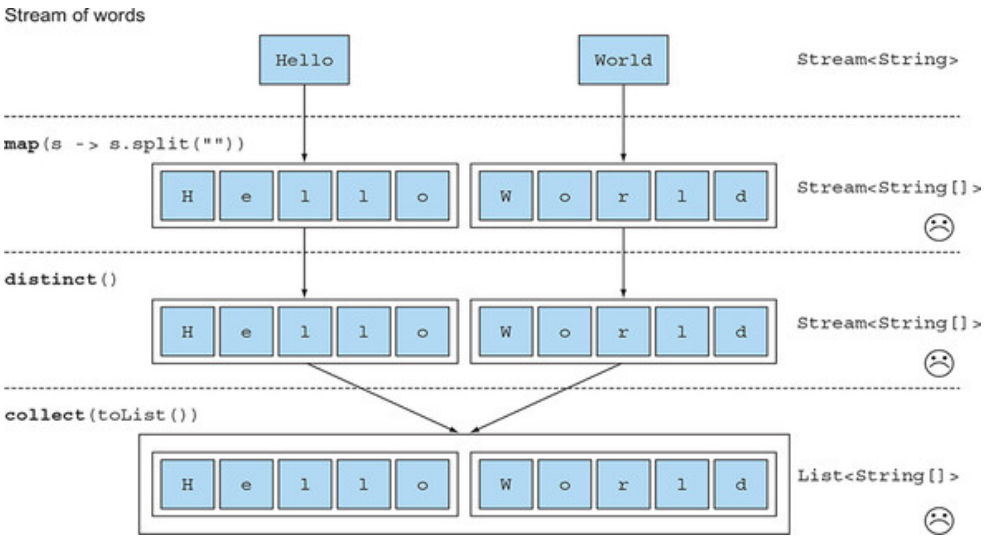
You saw how to return the length for each word in a list using the `map` method. Let's extend this idea a bit further: How could you return a list of all the *unique characters* for a list of words? For example, given the list of words `["Hello," "World"]` you'd like to return the list `["H," "e," "l," "o," "W," "r," "d"]`.

You might think that this is easy, that you can map each word into a list of characters and then call `distinct` to filter duplicate characters. A first go could be like the following:

```
words.stream()
     .map(word -> word.split(""))
     .distinct()
     .collect(toList());
```

The problem with this approach is that the lambda passed to the `map` method returns a `String[]` (an array of String) for each word. The stream returned by the `map` method is of type `Stream<String[]>`. What you want is `Stream<String>` to represent a stream of characters. **Figure 5.5** illustrates the problem.

Figure 5.5. Incorrect use of `map` to find unique characters from a list of words



Luckily there's a solution to this problem using the method `flatMap`! ...lve it.

...ys.stream

First, you need a stream of characters instead of a stream of arrays. There's a method called `Arrays.stream()` that takes an array and produces a stream:

```
String[] arrayOfWords = {"Goodbye", "World"};
Stream<String> streamOfwords = Arrays.stream(arrayOfWords);
```

Use it in the previous pipeline to see what happens:

```
words.stream()
     .map(word -> word.split(""))          1
     .map(Arrays::stream)                  2
     .distinct()
     .collect(toList());
```

- ***1* Converts each word into an array of its individual letters**
- ***2* Makes each array into a separate stream**

The current solution still doesn't work! This is because you now end up with a list of streams (more precisely, `List<Stream<String>>`). Indeed, you first convert each word into an array of its individual letters and then make each array into a separate stream.

Using flatMap

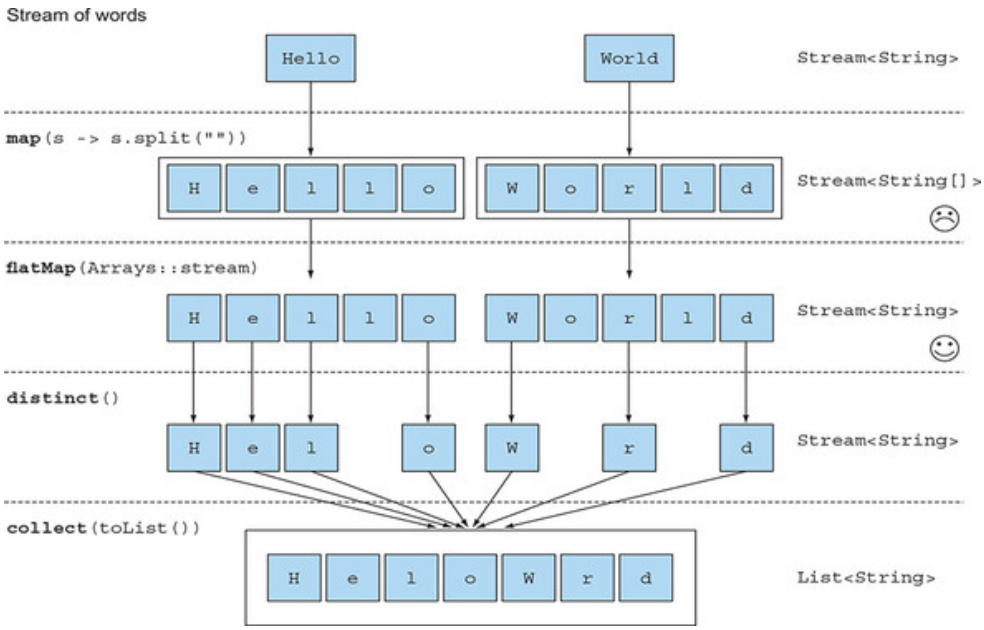
You can fix this problem by using `flatMap` as follows:

```
List<String> uniqueCharacters =
  words.stream()
       .map(word -> word.split(""))        1
       .flatMap(Arrays::stream)            2
       .distinct()
       .collect(toList());
```

- ***1* Converts each word into an array of its individual letters**
- ***2* Flattens each generated stream into a single stream**

Using the `flatMap` method has the effect of mapping each array not with a stream but *with the contents of that stream*. All the separate streams that were generated when using `map(Arrays::stream)` get amalgamated—flattened into a single stream. **Figure 5.6** illustrates the effect of using the `flatMap` method. Compare it with what `map` does in **figure 5.5**.

find the unique characters from a list of



In a nutshell, the `flatMap` method lets you replace each value of a stream with another stream and then concatenates all the generated streams into a single stream.

We'll come back to `flatMap` in **chapter 11** when we discuss more advanced Java 8 patterns such as using the new library class `Optional` for `null` checking. To solidify your understanding of `map` and `flatMap`, try out quiz 5.2.

Quiz 5.2: Mapping

1. Given a list of numbers, how would you return a list of the square of each number? For example, given [1, 2, 3, 4, 5] you should return [1, 4, 9, 16, 25].

Answer:

You can solve this problem by using `map` with a lambda that takes a number and returns the square of the number:

```
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5);
List<Integer> squares =
    numbers.stream()
           .map(n -> n * n)
           .List());
```


how would you return all pairs of numbers? For example, given a list [1, 2, 3] and a list [3, 4] you should return [(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4)]. For simplicity, you can represent a pair as an array with two elements.

Answer:

You could use two `maps` to iterate on the two lists and generate the pairs. But this would return a `Stream<Stream<Integer[]>>`. What you need to do is flatten the generated streams to result in a `Stream<Integer[]>`. This is what `flatMap` is for:

```
List<Integer> numbers1 = Arrays.asList(1, 2, 3);
List<Integer> numbers2 = Arrays.asList(3, 4);
List<int[]> pairs =
    numbers1.stream()
            .flatMap(i -> numbers2.stream()
                                  .map(j -> new int[]{i, j})
                    )
            .collect(toList());
```

3. How would you extend the previous example to return only pairs whose sum is divisible by 3?

Answer:

You saw earlier that `filter` can be used with a predicate to filter elements from a stream. Because after the `flatMap` operation you have a stream of `int[]` that represent a pair, you only need a predicate to check if the sum is divisible by 3:

```
List<Integer> numbers1 = Arrays.asList(1, 2, 3);
List<Integer> numbers2 = Arrays.asList(3, 4);
List<int[]> pairs =
    numbers1.stream()
            .flatMap(i ->
                     numbers2.stream()
                             .filter(j -> (i + j) % 3 == 0)
                             .map(j -> new int[]{i, j})
                    )
            .collect(toList());
```

The result is [(2, 4), (3, 3)].

Another common data processing idiom is finding whether some elements in a set of data match a given property. The Streams API provides such facilities through the `allMatch`, `anyMatch`, `noneMatch`, `findFirst`, and `findAny` methods of a stream.

5.4.1. Checking to see if a predicate matches at least one element

The `anyMatch` method can be used to answer the question "Is there an element in the stream matching the given predicate?" For example, you can use it to find out whether the menu has a vegetarian option:

```
if(menu.stream().anyMatch(Dish::isVegetarian)) {
    System.out.println("The menu is (somewhat) vegetarian friendly!!")
}
```

The `anyMatch` method returns a `boolean` and is therefore a terminal operation.

5.4.2. Checking to see if a predicate matches all elements

The `allMatch` method works similarly to `anyMatch` but will check to see if all the elements of the stream match the given predicate. For example, you can use it to find out whether the menu is healthy (all dishes are below 1000 calories):

```
boolean isHealthy = menu.stream()
                        .allMatch(dish -> dish.getCalories() < 1000);
```

noneMatch

The opposite of `allMatch` is `noneMatch`. It ensures that no elements in the stream match the given predicate. For example, you could rewrite the previous example as follows using `noneMatch`:

```
boolean isHealthy = menu.stream()
                        .noneMatch(d -> d.getCalories() >= 1000);
```

These three operations—`anyMatch`, `allMatch`, and `noneMatch`—make use of what we call *short-circuiting*, a stream version of the familiar Java short-circuiting && and || operators.

process the whole stream to produce a re-
d to evaluate a large `boolean` expression chained with `and` operators. You need only find out that one expression is `false` to deduce that the whole expression will return `false`, no matter how long the expression is; there's no need to evaluate the entire expression. This is what *short-circuiting* refers to.

In relation to streams, certain operations such as `allMatch`, `noneMatch`, `findFirst`, and `findAny` don't need to process the whole stream to produce a result. As soon as an element is found, a result can be produced. Similarly, `limit` is also a short-circuiting operation. The operation only needs to create a stream of a given size without processing all the elements in the stream. Such operations are useful (for example, when you need to deal with streams of infinite size, because they can turn an infinite stream into a stream of finite size). We'll show examples of infinite streams in **section 5.7**.

5.4.3. Finding an element

The `findAny` method returns an arbitrary element of the current stream. It can be used in conjunction with other stream operations. For example, you may wish to find a dish that's vegetarian. You can combine the `filter` method and `findAny` to express this query:

```
Optional<Dish> dish =
  menu.stream()
      .filter(Dish::isVegetarian)
      .findAny();
```

The stream pipeline will be optimized behind the scenes to perform a single pass and finish as soon as a result is found by using short-circuiting. But wait a minute; what's this `Optional` thing in the code?

Optional in a nutshell

The `Optional<T>` class (`java.util.Optional`) is a container class to represent the existence or absence of a value. In the previous code, it's possible that `findAny` doesn't find any element. Instead of returning `null`, which is well known for being error-prone, the Java 8 library designers introduced `Optional<T>`. We won't go into the details of `Optional` here, because we'll show in detail in **chapter 11** how your code can benefit from using `Optional` to avoid bugs related to `null` checking. But for now, it's good to know that there are a few methods available in `Optional` to explicitly check for the presence of a value or deal with the absence of a value:

- `isPresent()` returns `true` if `Optional` contains a value, `false` otherwise.
- `ifPresent(Consumer<T> block)` executes the given block if a value is present. We introduced the `Consumer` functional interface in **chap-**

ter 3; it lets you pass a lambda that takes an argument of type `T` and returns `void`.
- `T get()` returns the value if present; otherwise it throws a `NoSuchElement-Exception`.
- `T orElse(T other)` returns the value if present; otherwise it returns a default value.

For example, in the previous code you'd need to explicitly check for the presence of a dish in the `Optional` object to access its name:

```
menu.stream()
    .filter(Dish::isVegetarian)
    .findAny()                                                    1
    .ifPresent(dish -> System.out.println(dish.getName());        2
```

- ***1*** **Returns an Optional<Dish>.**
- ***2*** **If a value is contained, it's printed; otherwise nothing happens.**

5.4.4. Finding the first element

Some streams have an *encounter order* that specifies the order in which items logically appear in the stream (for example, a stream generated from a `List` or from a sorted sequence of data). For such streams you may wish to find the first element. There's the `findFirst` method for this, which works similarly to `findAny` (for example, the code that follows, given a list of numbers, finds the first square that's divisible by 3):

```
List<Integer> someNumbers = Arrays.asList(1, 2, 3, 4, 5);
Optional<Integer> firstSquareDivisibleByThree =
  someNumbers.stream()
             .map(n -> n * n)
             .filter(n -> n % 3 == 0)
             .findFirst(); // 9
```

When to use findFirst and findAny

You may wonder why we have both `findFirst` and `findAny`. The an- e first element is more constraining in par- hich element is returned, use `findAny` be- n using parallel streams.

5.5. Reducing

The terminal operations you've seen either return a `boolean` (`allMatch` and so on), `void` (`forEach`), or an `Optional` object (`findAny` and so on). You've also been using `collect` to combine all elements in a stream into a `List`.

In this section, you'll see how you can combine elements of a stream to express more complicated queries such as "Calculate the sum of all calories in the menu," or "What is the highest calorie dish in the menu?" using the `reduce` operation. Such queries combine all the elements in the stream repeatedly to produce a single value such as an `Integer`. These queries can be classified as *reduction operations* (a stream is reduced to a value). In functional programming-language jargon, this is referred to as a *fold* because you can view this operation as repeatedly folding a long piece of paper (your stream) until it forms a small square, which is the result of the fold operation.

5.5.1. Summing the elements

Before we investigate how to use the `reduce` method, it helps to first see how you'd sum the elements of a list of numbers using a `for-each` loop:

```
int sum = 0;
for (int x : numbers) {
    sum += x;
}
```

Each element of `numbers` is combined iteratively with the addition operator to form a result. You *reduce* the list of numbers into one number by repeatedly using addition. There are two parameters in this code:

- The initial value of the sum variable, in this case `0`
- The operation to combine all the elements of the list, in this case +

Wouldn't it be great if you could also multiply all the numbers without having to repeatedly copy and paste this code? This is where the `reduce` operation, which abstracts over this pattern of repeated application, can help. You can sum all the elements of a stream as follows:

```
eam().reduce(0, (a, b) -> a + b);
```

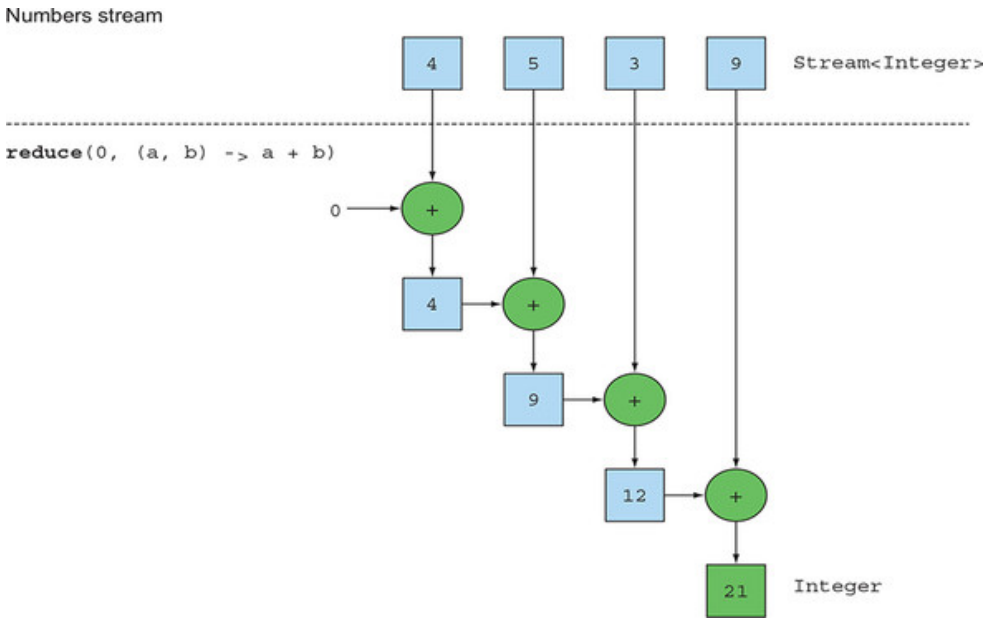

- An initial value, here `0`.
- A `BinaryOperator<T>` to combine two elements and produce a new value; here you use the lambda `(a, b) -> a + b`.

You could just as easily multiply all the elements by passing a different lambda, `(a, b) -> a * b`, to the `reduce` operation:

```
int product = numbers.stream().reduce(1, (a, b) -> a * b);
```

Figure 5.7 illustrates how the `reduce` operation works on a stream: the lambda combines each element repeatedly until the stream containing the integers 4, 5, 3, 9, are reduced to a single value.

Figure 5.7. Using `reduce` to sum the numbers in a stream



Let's take an in-depth look into how the `reduce` operation happens to sum a stream of numbers. First, `0` is used as the first parameter of the lambda (`a`), and 4 is consumed from the stream and used as the second parameter (`b`). `0 + 4` produces 4, and it becomes the new accumulated value. Then the lambda is called again with the accumulated value and the next element of the stream, 5, which produces the new accumulated value, 9. Moving forward, the lambda is called again with the accumulated value and the next element, 3, which produces 12. Finally, the lambda is called with 12 and the last element of the stream, 9, which produces the final value, 21.

You can make this code more concise by using a method reference. From Java 8 the `Integer` class now comes with a static `sum` method to add two ant instead of repeatedly writing out the


```
int sum = numbers.stream().reduce(0, Integer::sum);
```

No initial value

There's also an overloaded variant of `reduce` that doesn't take an initial value, but it returns an `Optional` object:

```
Optional<Integer> sum = numbers.stream().reduce((a, b) -> (a + b));
```

Why does it return an `Optional<Integer>`? Consider the case when the stream contains no elements. The `reduce` operation can't return a sum because it doesn't have an initial value. This is why the result is wrapped in an `Optional` object to indicate that the sum may be absent. Now see what else you can do with `reduce`.

5.5.2. Maximum and minimum

It turns out that reduction is all you need to compute maxima and minima as well! Let's see how you can apply what you just learned about `reduce` to calculate the maximum or minimum element in a stream. As you saw, `reduce` takes two parameters:

- An initial value
- A lambda to combine two stream elements and produce a new value

The lambda is applied step-by-step to each element of the stream with the addition operator, as shown in **figure 5.7**. You need a lambda that, given two elements, returns the maximum of them. The `reduce` operation will use the new value with the next element of the stream to produce a new maximum until the whole stream is consumed! You can use `reduce` as follows to calculate the maximum in a stream; this is illustrated in **figure 5.8**.

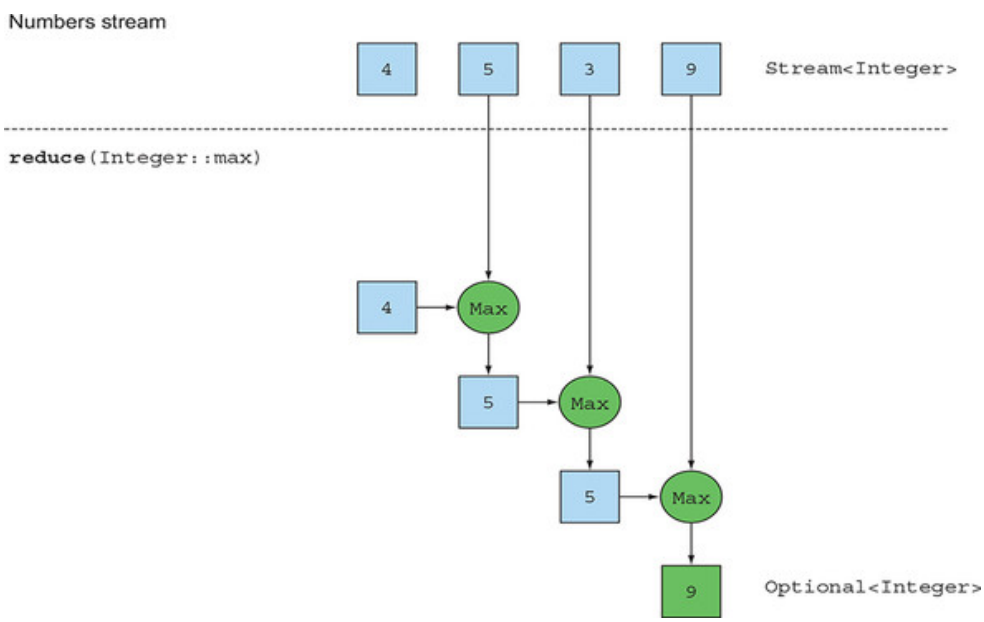
```
Optional<Integer> max = numbers.stream().reduce(Integer::max);
```

To calculate the minimum, you need to pass `Integer.min` to the `reduce` operation instead of `Integer.max`:

```
Optional<Integer> min = numbers.stream().reduce(Integer::min);
```


ed the lambda `(x, y) -> x < y ? x :`
ut the latter is definitely easier to read!

n—calculating the maximum



To test your understanding of the `reduce` operation, try quiz 5.3.

Quiz 5.3: Reducing
How would you count the number of dishes in a stream using the `map` and `reduce` methods?

Answer:

You can solve this problem by mapping each element of a stream into the number 1 and then summing them using `reduce`! This is equivalent to counting, in order, the number of elements in the stream:

```
int count = menu.stream()
                .map(d -> 1)
                .reduce(0, (a, b) -> a + b);
```

A chain of `map` and `reduce` is commonly known as the map-reduce pattern, made famous by Google's use of it for web searching because it can be easily parallelized. Note that in **chapter 4** you saw the built-in method `count` to count the number of elements in the stream:

```
long count = menu.stream().count();
```


Benefit of the reduce method and parallelism

The benefit of using `reduce` compared to the step-by-step iteration summation that you wrote earlier is that the iteration is abstracted using internal iteration, which enables the internal implementation to choose to perform the `reduce` operation in parallel. The iterative summation example involves shared updates to a `sum` variable, which doesn't parallelize gracefully. If you add in the needed synchronization, you'll likely discover that thread contention robs you of all the performance that parallelism was supposed to give you! Parallelizing this computation requires a different approach: partition the input, sum the partitions, and combine the sums. But now the code is starting to look very different. You'll see what this looks like in **chapter 7** using the fork/join framework. But for now it's important to realize that the mutable-accumulator pattern is a dead end for parallelization. You need a new pattern, and this is what `reduce` provides you. You'll also see in **chapter 7** that to sum all the elements in parallel using streams, there's almost no modification to your code: `stream()` becomes `parallelStream()`:

```
int sum = numbers.parallelStream().reduce(0, Integer::sum);
```

But there's a price to pay to execute this code in parallel, as we'll explain later: the lambda passed to `reduce` can't change state (for example, instance variables), and the operation needs to be associative and commutative so it can be executed in any order.

You have seen reduction examples that produced an `Integer`: the sum of a stream, the maximum of a stream, or the number of elements in a stream. You'll see, in **section 5.6**, that additional built-in methods such as `sum` and `max` are available to help you write slightly more concise code for common reduction patterns. We'll investigate a more complex form of reductions using the `collect` method in the next chapter. For example, instead of reducing a stream into an `Integer`, you can also reduce it into a `Map` if you want to group dishes by types.

Stream operations: stateless vs. stateful

erations. An initial presentation can make
ing works smoothly, and you get par-
`parallelStream` instead of `stream` to get a stream from a collection.

Certainly for many applications this is the case, as you've seen in the previous examples. You can turn a list of dishes into a stream, `filter` to se-

lect various dishes of a certain type, then `map` down the resulting stream to add on the number of calories, and then `reduce` to produce the total number of calories of the menu. You can even do such stream calculations in parallel. But these operations have different characteristics. There are issues about what internal state they need to operate.

Operations like `map` and `filter` take *each* element from the input stream and produce *zero or one* result in the output stream. These operations are in general *stateless*: they don't have an internal state (assuming the user-supplied lambda or method reference has no internal mutable state).

But operations like `reduce`, `sum`, and `max` need to have internal state to accumulate the result. In this case the internal state is small. In our example it consisted of an `int` or `double`. The internal state is of *bounded size* no matter how many elements are in the stream being processed.

By contrast, some operations such as `sorted` or `distinct` seem at first to behave like `filter` or `map`—all take a stream and produce another stream (an intermediate operation)—but there's a crucial difference. Both sorting and removing duplicates from a stream require knowing the previous history to do their job. For example, sorting requires *all the elements to be buffered* before a single item can be added to the output stream; the storage requirement of the operation is *unbounded*. This can be problematic if the data stream is large or infinite. (What should reversing the stream of all prime numbers do? It should return the largest prime number, which mathematics tells us doesn't exist.) We call these operations *stateful operations*.

You've now seen a lot of stream operations that you can use to express sophisticated data processing queries! **Table 5.1** summarizes the operations seen so far. You get to practice them in the next section through an exercise.

Table 5.1. Intermediate and terminal operations

Operation	Type	Return type	Type/functional interface used	Function descriptor
		Stream<T>	Predicate<T>	T -> boolean

Operation	Type	Return type	Type/functional interface used	Function descriptor
distinct	Intermediate (stateful-unbounded)	Stream<T>		
takeWhile	Intermediate	Stream<T>	Predicate<T>	T -> boolean
dropWhile	Intermediate	Stream<T>	Predicate<T>	T -> boolean
skip	Intermediate (stateful-bounded)	Stream<T>	long	
limit	Intermediate (stateful-bounded)	Stream<T>	long	
map	Intermediate	Stream<R>	Function<T, R>	T -> R
flatMap	Intermediate	Stream<R>	Function<T, Stream<R>>	T -> Stream<R>
sorted	Intermediate (stateful-unbounded)	Stream<T>	Comparator<T>	(T, T) -> int
anyMatch	Terminal	boolean	Predicate<T>	T -> boolean
noneMatch	Terminal	boolean	Predicate<T>	T -> boolean
allMatch	Terminal	boolean	Predicate<T>	T -> boolean
		Optional<T>		
findFirst	Terminal	Optional<T>		
forEach	Terminal	void	Consumer<T>	T -> void

Operation	Type	Return type	Type/functional interface used	Function descriptor
collect	Terminal	R	Collector<T, A, R>	
reduce	Terminal (stateful-bounded)	Optional<T>	BinaryOperator<T>	(T, T) -> T
count	Terminal	long		

5.6. Putting it all into practice

In this section, you get to practice what you've learned about streams so far. We now explore a different domain: traders executing transactions. You're asked by your manager to find answers to eight queries. Can you do it? We give the solutions in **section 5.6.2**, but you should try them yourself first to get some practice:

1. Find all transactions in the year 2011 and sort them by value (small to high).
2. What are all the unique cities where the traders work?
3. Find all traders from Cambridge and sort them by name.
4. Return a string of all traders' names sorted alphabetically.
5. Are any traders based in Milan?
6. Print the values of all transactions from the traders living in Cambridge.
7. What's the highest value of all the transactions?
8. Find the transaction with the smallest value.

5.6.1. The domain: Traders and Transactions

Here's the domain you'll be working with, a list of `Traders` and `Transactions`:

```
Trader raoul = new Trader("Raoul", "Cambridge");
Trader mario = new Trader("Mario","Milan");
Trader alan = new Trader("Alan","Cambridge");
                                    ader("Brian","Cambridge");
                                    nsactions = Arrays.asList(
                                    rian, 2011, 300),
                                    aoul, 2012, 1000),
    new Transaction(raoul, 2011, 400),
    new Transaction(mario, 2012, 710),
    new Transaction(mario, 2012, 700),
```

```
        new Transaction(alan, 2012, 950)
    );
```

`Trader` and `Transaction` are classes defined as follows:

```
public class Trader{
    private final String name;
    private final String city;
    public Trader(String n, String c){
        this.name = n;
        this.city = c;
    }
    public String getName(){
        return this.name;
    }
    public String getCity(){
        return this.city;
    }
    public String toString(){
        return "Trader:"+this.name + " in " + this.city;
    }
}
public class Transaction{
    private final Trader trader;
    private final int year;
    private final int value;
    public Transaction(Trader trader, int year, int value){
        this.trader = trader;
        this.year = year;
        this.value = value;
    }
    public Trader getTrader(){
        return this.trader;
    }
    public int getYear(){
        return this.year;
    }
    public int getValue(){
        return this.value;
    }
    public String toString(){
        return "{" + this.trader + ", " +
               "year: "+this.year+", " +
               " + this.value +"}";
```


5.6.2. Solutions

We now provide the solutions in the following code listings, so you can verify your understanding of what you've learned so far. Well done!

Listing 5.1. Finds all transactions in 2011 and sort by value (small to high)

```
List<Transaction> tr2011 =
    transactions.stream()
                .filter(transaction -> transaction.getYear() == 2011)
                .sorted(comparing(Transaction::getValue))
                .collect(toList());
```

- ***1*** **Passes a predicate to filter to select transactions in year 2011**
- ***2*** **Sorts them by using the value of the transaction**
- ***3*** **Collects all the elements of the resulting Stream into a List**

Listing 5.2. What are all the unique cities where the traders work?

```
List<String> cities =
    transactions.stream()
                .map(transaction -> transaction.getTrader().getCity())
                .distinct()
                .collect(toList());
```

- ***1*** **Extracts the city from each trader associated with the transaction**
- ***2*** **Selects only unique cities**

You haven't seen this yet, but you could also drop `distinct()` and use `toSet()` instead, which would convert the stream into a set. You'll learn more about it in **chapter 6**.

```
Set<String> cities =
    transactions.stream()
                .map(transaction -> transaction.getTrader().getCity())
                .collect(toSet());
```

Listing 5.3. Finds all traders from Cambridge and sort them by name


```
                .m()
                ransaction::getTrader)
                .filter(trader -> trader.getCity().equals("Cambridge")
                .distinct()
                .sorted(comparing(Trader::getName))
                .collect(toList());
```

- ***1*** **Extracts all traders from the transactions**
- ***2*** **Selects only the traders from Cambridge**
- ***3*** **Removes any duplicates**
- ***4*** **Sorts the resulting stream of traders by their names**

Listing 5.4. Returns a string of all traders' names sorted alphabetically

```
String traderStr =
    transactions.stream()
                .map(transaction -> transaction.getTrader().getName())
                .distinct()
                .sorted()
                .reduce("", (n1, n2) -> n1 + n2);
```

- ***1*** **Extracts all the names of the traders as a Stream of Strings**
- ***2*** **Removes duplicate names**
- ***3*** **Sorts the names alphabetically**
- ***4*** **Combines the names one by one to form a String that concatenates all the names**

Note that this solution is inefficient (all Strings are repeatedly concatenated, which creates a new `String` object at each iteration). In the next chapter, you'll see a more efficient solution that uses `joining()` as follows (which internally makes use of a `StringBuilder`):

```
String traderStr =
    transactions.stream()
                .map(transaction -> transaction.getTrader().getName())
                .distinct()
                .sorted()
                .collect(joining());
```

Listing 5.5. Are any traders based in Milan?

```
boolean milanBased =
    transactions.stream()
                .anyMatch(transaction -> transaction.getTrader()
                                                    .getCity()
                                                    .equals("Milan"));
```


latch to check if there's a trader from Milan.

Listing 5.6. Prints all transactions' values from the traders living in Cambridge

```
transactions.stream()
            .filter(t -> "Cambridge".equals(t.getTrader().getCity()))
            .map(Transaction::getValue)
            .forEach(System.out::println);
```

- ***1*** **Selects the transactions where the traders live in Cambridge**
- ***2*** **Extracts the values of these trades**
- ***3*** **Prints each value**

Listing 5.7. What's the highest value of all the transactions?

```
Optional<Integer> highestValue =
    transactions.stream()
                .map(Transaction::getValue)          1
                .reduce(Integer::max);               2
```

- ***1*** **Extracts the value of each transaction**
- ***2*** **Calculates the max of the resulting stream**

Listing 5.8. Finds the transaction with the smallest value

```
Optional<Transaction> smallestTransaction =
  transactions.stream()
              .reduce((t1, t2) ->
                        t1.getValue() < t2.getValue() ? t1 : t2);
```

- ***1*** **Finds the smallest transaction by repeatedly comparing the values of each transaction**

You can do better. A stream supports the methods `min` and `max` that take a `Comparator` as argument to specify which key to compare with when calculating the minimum or maximum:

```
Optional<Transaction> smallestTransaction =
  transactions.stream()
              .min(comparing(Transaction::getValue));
```


use the `reduce` method to calculate the
n. For example, you can calculate the number of calories in the menu as follows:

```
int calories = menu.stream()
                   .map(Dish::getCalories)
```

```
                .reduce(0, Integer::sum);
```

The problem with this code is that there's an insidious boxing cost. Behind the scenes each `Integer` needs to be unboxed to a primitive before performing the summation. In addition, wouldn't it be nicer if you could call a `sum` method directly as follows?

```
int calories = menu.stream()
                   .map(Dish::getCalories)
                   .sum();
```

But this isn't possible. The problem is that the method `map` generates a `Stream<T>`. Even though the elements of the stream are of type `Integer`, the `streams` interface doesn't define a `sum` method. Why not? Say you had only a `Stream<Dish>` like the `menu`; it wouldn't make any sense to be able to sum dishes. But don't worry; the Streams API also supplies *primitive stream specializations* that support specialized methods to work with streams of numbers.

5.7.1. Primitive stream specializations

Java 8 introduces three primitive specialized stream interfaces to tackle this issue, `IntStream`, `DoubleStream`, and `LongStream`, which respectively specialize the elements of a stream to be `int`, `long`, and `double`—and thereby avoid hidden boxing costs. Each of these interfaces brings new methods to perform common numeric reductions, such as `sum` to calculate the sum of a numeric stream and `max` to find the maximum element. In addition, they have methods to convert back to a stream of objects when necessary. The thing to remember is that the additional complexity of these specializations isn't inherent to streams. It reflects the complexity of boxing—the (efficiency-based) difference between `int` and `Integer` and so on.

Mapping to a numeric stream

The most common methods you'll use to convert a stream to a specialized version are `mapToInt`, `mapToDouble`, and `mapToLong`. These methods work exactly like the method `map` that you saw earlier but return a specialized stream instead of a `Stream<T>`. For example, you can use `mapToInt` to calculate the sum of calories in the `menu`:


```
int calories = menu.stream()                          1
                   .mapToInt(Dish::getCalories)       2
                   .sum();
```

- *1* **Returns a Stream<Dish>**
- *2* **Returns an IntStream**

Here, the method `mapToInt` extracts all the calories from each dish (represented as an `Integer`) and returns an `IntStream` as the result (rather than a `Stream<Integer>`). You can then call the `sum` method defined on the `IntStream` interface to calculate the sum of calories! Note that if the stream were empty, `sum` would return `0` by default. `IntStream` also supports other convenience methods such as `max`, `min`, and `average`.

Converting back to a stream of objects

Similarly, once you have a numeric stream, you may be interested in converting it back to a nonspecialized stream. For example, the operations of an `IntStream` are restricted to produce primitive integers: the `map` operation of an `IntStream` takes a lambda that takes an `int` and produces an `int` (an `IntUnaryOperator`). But you may want to produce a different value such as a `Dish`. For this you need to access the operations defined in the `Streams` interface that are more general. To convert from a primitive stream to a general stream (each `int` will be boxed to an `Integer`) you can use the method `boxed`, as follows:

```
IntStream intStream = menu.stream().mapToInt(Dish::getCalories);      1
Stream<Integer> stream = intStream.boxed();                           2
```

- *1* **Converts a Stream to a numeric stream**
- *2* **Converts the numeric stream to a Stream**

You'll learn in the next section that `boxed` is particularly useful when you deal with numeric ranges that need to be boxed into a general stream.

Default values: OptionalInt

The sum example was convenient because it has a default value: `0`. But if you want to calculate the maximum element in an `IntStream`, you'll need something different because `0` is a wrong result. How can you differentiate that the stream has no element and that the real maximum is `0`? Earlier we introduced the `Optional` class, which is a container that indi- of a value. `Optional` can be parameterized nteger, `String`, and so on. There's a prim- ional as well for the three primitive stream specializations: `OptionalInt`, `OptionalDouble`, and `OptionalLong`.

For example, you can find the maximal element of an `IntStream` by calling the `max` method, which returns an `OptionalInt`:

```
OptionalInt maxCalories = menu.stream()
                               .mapToInt(Dish::getCalories)
                               .max();
```

You can now process the `OptionalInt` explicitly to define a default value if there's no maximum:

```
int max = maxCalories.orElse(1);          1
```

- ***1*** **Provides an explicit default maximum if there's no value**

5.7.2. Numeric ranges

A common use case when dealing with numbers is working with ranges of numeric values. For example, suppose you'd like to generate all numbers between 1 and 100. Java 8 introduces two static methods available on `IntStream` and `LongStream` to help generate such ranges: `range` and `rangeClosed`. Both methods take the starting value of the range as the first parameter and the end value of the range as the second parameter. But `range` is exclusive, whereas `rangeClosed` is inclusive. Let's look at an example:

```
IntStream evenNumbers = IntStream.rangeClosed(1, 100)          1
                                 .filter(n -> n % 2 == 0);     2
System.out.println(evenNumbers.count());                       3
```

- ***1*** **Represents the range 1 to 100**
- ***2*** **Represents stream of even numbers from 1 to 100**
- ***3*** **Represents 50 even numbers from 1 to 100**

Here you use the `rangeClosed` method to generate a range of all numbers from 1 to 100. It produces a stream so you can chain the `filter` method to select only even numbers. At this stage no computation has been done. Finally, you call `count` on the resulting stream. Because `count` process the stream and return the result n numbers from 1 to 100, inclusive. Note e using `IntStream.range(1, 100)` instead, the result would be 49 even numbers because `range` is exclusive.

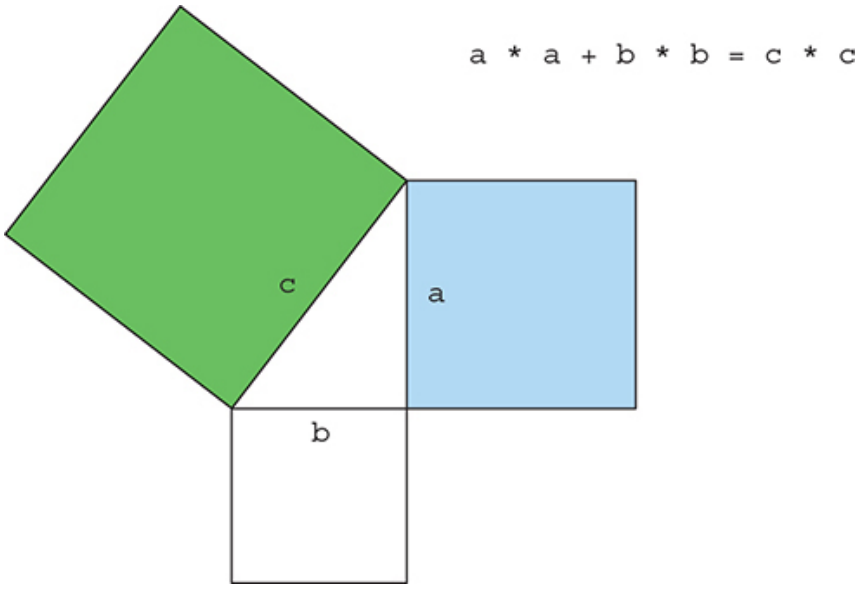
5.7.3. Putting numerical streams into practice: Pythagorean triples

Now we'll look at a more difficult example so you can solidify what you've learned about numeric streams and all the stream operations you've learned so far. Your mission, if you choose to accept it, is to create a stream of Pythagorean triples.

Pythagorean triple

What's a Pythagorean triple? We have to go back a few years in the past. In one of your exciting math classes, you learned that the famous Greek mathematician Pythagoras discovered that certain triples of numbers `(a, b, c)` satisfy the formula `a * a + b * b = c * c` where `a`, `b`, and `c` are integers. For example, (3, 4, 5) is a valid Pythagorean triple because 3 * 3 + 4 * 4 = 5 * 5 or 9 + 16 = 25. There are an infinite number of such triples. For example, (5, 12, 13), (6, 8, 10), and (7, 24, 25) are all valid Pythagorean triples. Such triples are useful because they describe the three side lengths of a right-angled triangle, as illustrated in **figure 5.9**.

Figure 5.9. The Pythagorean theorem



Representing a triple

Where do you start? The first step is to define a triple. Instead of (more properly) defining a new class to represent a triple, you can use an array of `int` with three elements. For example, `new int[]{3, 4, 5}` to represent the tuple (3, 4, 5). You can now access each individual component of the tuple using array indexing.

s you with the first two numbers of the triple: `a` and `b`. How do you know whether that will form a good combination? You need to test whether the square root of `a * a + b * b` is a whole number. This is expressed in Java as `Math.sqrt(a*a + b*b) % 1 == 0`. (Given a floating-point number, x, in Java its fractional part is ob-

tained using x `% 1.0`, and whole numbers like 5.0 have zero fractional part.) Our code uses this idea in a `filter` operation (you'll see how to use this later to form valid code):

```
filter(b -> Math.sqrt(a*a + b*b) % 1 == 0)
```

Assuming that surrounding code has given a value for `a`, and assuming `stream` provides possible values for `b`, `filter` will select only those values for `b` that can form a Pythagorean triple with `a`.

Generating tuples

Following the `filter`, you know that both `a` and `b` can form a correct combination. You now need to create a triple. You can use the `map` operation to transform each element into a Pythagorean triple as follows:

```
stream.filter(b -> Math.sqrt(a*a + b*b) % 1 == 0)
      .map(b -> new int[]{a, b, (int) Math.sqrt(a * a + b * b)});
```

Generating b values

You're getting closer! You now need to generate values for `b`. You saw that `Stream .rangeClosed` allows you to generate a stream of numbers in a given interval. You can use it to provide numeric values for `b`, here 1 to 100:

```
IntStream.rangeClosed(1, 100)
         .filter(b -> Math.sqrt(a*a + b*b) % 1 == 0)
         .boxed()
         .map(b -> new int[]{a, b, (int) Math.sqrt(a * a + b * b)});
```

Note that you call `boxed` after the `filter` to generate a `Stream<Integer>` from the `IntStream` returned by `rangeClosed`. This is because `map` returns an array of `int` for each element of the stream. The `map` method from an `IntStream` expects only another `int` to be returned for each element of the stream, which isn't what you want! You can rewrite this using the method `mapToObj` of an `IntStream`, which re-


```
1, 100)
         .filter(b -> Math.sqrt(a*a + b*b) % 1 == 0)
         .mapToObj(b -> new int[]{a, b, (int) Math.sqrt(a * a + b * b)
```

Generating a values

There's one crucial component that we assumed was given: the value for `a`. You now have a stream that produces Pythagorean triples provided the value `a` is known. How can you fix this? Just like with `b`, you need to generate numeric values for `a`! The final solution is as follows:

```
Stream<int[]> pythagoreanTriples =
    IntStream.rangeClosed(1, 100).boxed()
             .flatMap(a ->
                IntStream.rangeClosed(a, 100)
                         .filter(b -> Math.sqrt(a*a + b*b) % 1 == 0)
                         .mapToObj(b ->
                            new int[]{a, b, (int)Math.sqrt(a * a + b *
                     );
```

Okay, what's the `flatMap` about? First, you create a numeric range from 1 to 100 to generate values for `a`. For each given value of `a` you're creating a stream of triples. Mapping a value of `a` to a stream of triples would result in a stream of streams! The `flatMap` method does the mapping and also flattens all the generated streams of triples into a single stream. As a result, you produce a stream of triples. Note also that you change the range of `b` to be `a` to 100. There's no need to start the range at the value 1 because this would create duplicate triples (for example, (3, 4, 5) and (4, 3, 5)).

Running the code

You can now run your solution and select explicitly how many triples you'd like to return from the generated stream using the `limit` operation that you saw earlier:

```
pythagoreanTriples.limit(5)
                  .forEach(t ->
                     System.out.println(t[0] + ", " + t[1] + ", " + t[
```

This will print

```
3, 4, 5
5, 12, 13
```


Can you do better?

The current solution isn't optimal because you calculate the square root twice. One possible way to make your code more compact is to generate all triples of the form `(a*a, b*b, a*a+b*b)` and then filter the ones that match your criteria:

```
Stream<double[]> pythagoreanTriples2 =
  IntStream.rangeClosed(1, 100).boxed()
           .flatMap(a ->
              IntStream.rangeClosed(a, 100)
                       .mapToObj(
                          b -> new double[]{a, b, Math.sqrt(a*a + b*b)
                       .filter(t -> t[2] % 1 == 0));
```

- *1* **Produces triples**
- *2* **The third element of the tuple must be a whole number.**

5.8. Building streams

Hopefully, by now you're convinced that streams are powerful and useful to express data-processing queries. You were able to get a stream from a collection using the `stream` method. In addition, we showed you how to create numerical streams from a range of numbers. But you can create streams in many more ways! This section shows how you can create a stream from a sequence of values, from an array, from a file, and even from a generative function to create infinite streams!

5.8.1. Streams from values

You can create a stream with explicit values by using the static method `Stream.of`, which can take any number of parameters. For example, in the following code you create a stream of strings directly using `Stream.of`. You then convert the strings to uppercase before printing them one by one:

```
Stream<String> stream = Stream.of("Modern ", "Java ", "In ", "Action")
stream.map(String::toUpperCase).forEach(System.out::println);
```

You can get an empty stream using the `empty` method as follows:


```
ream = Stream.empty();
```

5.8.2. Stream from nullable

In Java 9, a new method was added that lets you create a stream from a nullable object. After playing with streams, you may have encountered a

situation where you extracted an object that may be null and then needs to be converted into a stream (or an empty stream for null). For example, the method `System.getProperty` returns `null` if there is no property with the given key. To use it together with a stream, you'd need to explicitly check for null as follows:

```
String homeValue = System.getProperty("home");
Stream<String> homeValueStream
    = homeValue == null ? Stream.empty() : Stream.of(value);
```

Using `Stream.ofNullable` you can rewrite this code more simply:

```
Stream<String> homeValueStream
    = Stream.ofNullable(System.getProperty("home"));
```

This pattern can be particularly handy in conjunction with `flatMap` and a stream of values that may include nullable objects:

```
Stream<String> values =
   Stream.of("config", "home", "user")
         .flatMap(key -> Stream.ofNullable(System.getProperty(key)));
```

5.8.3. Streams from arrays

You can create a stream from an array using the static method `Arrays.stream`, which takes an array as parameter. For example, you can convert an array of primitive `int`s into an `IntStream` and then sum the `IntStream` to produce an `int`, as follows:

```
int[] numbers = {2, 3, 5, 7, 11, 13};
int sum = Arrays.stream(numbers).sum();          1
```

- ***1*** **The sum is 41.**

5.8.4. Streams from files

Java's NIO API (non-blocking I/O), which is used for I/O operations such as ...ated to take advantage of the Streams API. ...`nio.file.Files` return a stream. For ex... ...`es.lines`, which returns a stream of lines as strings from a given file. Using what you've learned so far, you could use this method to find out the number of unique words in a file as follows:

```
long uniqueWords = 0;
try(Stream<String> lines =
          Files.lines(Paths.get("data.txt"), Charset.defaultCharset())
uniqueWords = lines.flatMap(line -> Arrays.stream(line.split(" ")))
                   .distinct()
                   .count();
}
catch(IOException e){

}
```

- ***1*** **Streams are AutoCloseable so there's no need for try-finally**
- ***2*** **Generates a stream of words**
- ***3*** **Removes duplicates**
- ***4*** **Counts the number of unique words**
- ***5*** **Deals with the exception if one occurs when opening the file**

You use `Files.lines` to return a stream where each element is a line in the given file. This call is surrounded by a `try/catch` block because the source of the stream is an I/O resource. In fact, the call `Files.lines` will open an I/O resource, which needs to be closed to avoid a leak. In the past, you'd need an explicit `finally` block to do this. Conveniently, the `Stream` interface implements the interface `AutoCloseable`. This means that the management of the resource is handled for you within the `try` block. Once you have a stream of lines, you can then split each line into words by calling the `split` method on `line`. Notice how you use `flatMap` to produce one flattened stream of words instead of multiple streams of words for each line. Finally, you count each distinct word in the stream by chaining the methods `distinct` and `count`.

5.8.5. Streams from functions: creating infinite streams!

The Streams API provides two static methods to generate a stream from a function: `Stream.iterate` and `Stream.generate`. These two operations let you create what we call an *infinite stream*, a stream that doesn't have a fixed size like when you create a stream from a fixed collection. Streams produced by `iterate` and `generate` create values on demand given a function and can therefore calculate values forever! It's generally sensible to use `limit(n)` on such streams to avoid printing an infinite number

Let's look at a simple example of how to use `iterate` before we explain it:

```
Stream.iterate(0, n -> n + 2)
      .limit(10)
      .forEach(System.out::println);
```

The `iterate` method takes an initial value, here `0`, and a lambda (of type `Unary–Operator<T>`) to apply successively on each new value produced. Here you return the previous element added with 2 using the lambda `n -> n + 2`. As a result, the `iterate` method produces a stream of all even numbers: the first element of the stream is the initial value `0`. Then it adds 2 to produce the new value 2; it adds 2 again to produce the new value 4 and so on. This `iterate` operation is fundamentally sequential because the result depends on the previous application. Note that this operation produces an *infinite stream*—the stream doesn't have an end because values are computed on demand and can be computed forever. We say the stream is *unbounded*. As we discussed earlier, this is a key difference between a stream and a collection. You're using the `limit` method to explicitly limit the size of the stream. Here you select only the first 10 even numbers. You then call the `forEach` terminal operation to consume the stream and print each element individually.

In general, you should use `iterate` when you need to produce a sequence of successive values (for example, a date followed by its next date: January 31, February 1, and so on). To see a more difficult example of how you can apply `iterate`, try out quiz 5.4.

Quiz 5.4: Fibonacci tuples series

The Fibonacci series is famous as a classic programming exercise. The numbers in the following sequence are part of the Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55. . . . The first two numbers of the series are 0 and 1, and each subsequent number is the sum of the previous two.

The series of Fibonacci tuples is similar; you have a sequence of a number and its successor in the series: (0, 1), (1, 1), (1, 2), (2, 3), (3, 5), (5, 8), (8, 13), (13, 21). . . .

Your task is to generate the first 20 elements of the series of Fibonacci tu-

e first problem is that the `iterate` method takes a `UnaryOperator<T>` as argument, and you need a stream of tuples such as (0, 1). You can, again rather sloppily, use an array of two elements to represent a tuple. For example, `new int[]{0, 1}` represents the first

element of the Fibonacci series (0, 1). This will be the initial value of the `iterate` method:

```
Stream.iterate(new int[]{0, 1}, ???)
      .limit(20)
      .forEach(t -> System.out.println("(" + t[0] + "," + t[1] +")"));
```

In this quiz, you need to figure out the highlighted `???` in the code. Remember that `iterate` will apply the given lambda successively.

Answer:

```
Stream.iterate(new int[]{0, 1},
               t -> new int[]{t[1], t[0]+t[1]})
      .limit(20)
      .forEach(t -> System.out.println("(" + t[0] + "," + t[1] +")"));
```

How does it work? `iterate` needs a lambda to specify the successor element. In the case of the tuple (3, 5) the successor is (5, 3+5) = (5, 8). The next one is (8, 5+8). Can you see the pattern? Given a tuple, the successor is (t[1], t[0] + t[1]). This is what the following lambda specifies: `t -> new int[]{t[1],t[0] + t[1]}`. By running this code you'll get the series (0, 1), (1, 1), (1, 2), (2, 3), (3, 5), (5, 8), (8, 13), (13, 21). . . . Note that if you wanted to print the normal Fibonacci series, you could use a `map` to extract only the first element of each tuple:

```
Stream.iterate(new int[]{0, 1},
               t -> new int[]{t[1],t[0] + t[1]})
      .limit(10)
      .map(t -> t[0])
      .forEach(System.out::println);
```

This code will produce the Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34 . . .

In Java 9, the `iterate` method was enhanced with support for a predi-
erate numbers starting at 0 but stop the it-
ater than 100:


```
-> n < 100, n -> n + 4)
      .forEach(System.out::println);
```

The `iterate` method takes a predicate as its second argument that tells you when to continue iterating up until. Note that you may think that you can use the `filter` operation to achieve the same result:

```
IntStream.iterate(0, n -> n + 4)
         .filter(n -> n < 100)
         .forEach(System.out::println);
```

Unfortunately that isn't the case. In fact, this code wouldn't terminate! The reason is that there's no way to know in the filter that the numbers continue to increase, so it keeps on filtering them infinitely! You could solve the problem by using `takeWhile`, which would short-circuit the stream:

```
IntStream.iterate(0, n -> n + 4)
         .takeWhile(n -> n < 100)
         .forEach(System.out::println);
```

But, you have to admit that `iterate` with a predicate is a bit more concise!

Generate

Similarly to the method `iterate`, the method `generate` lets you produce an infinite stream of values computed on demand. But `generate` doesn't apply successively a function on each new produced value. It takes a lambda of type `Supplier<T>` to provide new values. Let's look at an example of how to use it:

```
Stream.generate(Math::random)
      .limit(5)
      .forEach(System.out::println);
```

This code will generate a stream of five random double numbers from 0 to 1. For example, one run gives the following:

```
0.9410810294106129
```


The static method `Math.random` is used as a generator for new values. Again you limit the size of the stream explicitly using the `limit` method;

otherwise the stream would be unbounded!

You may be wondering whether there's anything else useful you can do using the method `generate`. The supplier we used (a method reference to `Math.random`) was stateless: it wasn't recording any values somewhere that can be used in later computations. But a supplier doesn't have to be stateless. You can create a supplier that stores state that it can modify and use when generating the next value of the stream. As an example, we'll show how you can also create the Fibonacci series from quiz 5.4 using `generate` so that you can compare it with the approach using the `iterate` method! But it's important to note that a supplier that's stateful isn't safe to use in parallel code. The stateful `IntSupplier` for Fibonacci is shown at the end of this chapter for completeness but should generally be avoided! We discuss the problem of operations with side effects and parallel streams further in **chapter 7**.

We'll use an `IntStream` in our example to illustrate code that's designed to avoid boxing operations. The `generate` method on `IntStream` takes an `IntSupplier` instead of a `Supplier<T>`. For example, here's how to generate an infinite stream of ones:

```
IntStream ones = IntStream.generate(() -> 1);
```

You saw in **chapter 3** that lambdas let you create an instance of a functional interface by providing the implementation of the method directly inline. You can also pass an explicit object, as follows, by implementing the `getAsInt` method defined in the `IntSupplier` interface (although this seems gratuitously long-winded, please bear with us):

```
IntStream twos = IntStream.generate(new IntSupplier(){
            public int getAsInt(){
                return 2;
            }
        });
```

The `generate` method will use the given supplier and repeatedly call the `getAsInt` method, which always returns 2. But the difference between e and a lambda is that the anonymous class ch the `getAsInt` method can modify. This All lambdas you've seen so far were side-effect free, they didn't change any state.

To come back to our Fibonacci tasks, what you need to do now is create an `IntSupplier` that maintains in its state the previous value in the se-

ries, so `getAsInt` can use it to calculate the next element. In addition, it can update the state of the `IntSupplier` for the next time it's called. The following code shows how to create an `IntSupplier` that will return the next Fibonacci element when it's called:

```
IntSupplier fib = new IntSupplier(){
    private int previous = 0;
    private int current = 1;
    public int getAsInt(){
        int oldPrevious = this.previous;
        int nextValue = this.previous + this.current;
        this.previous = this.current;
        this.current = nextValue;
        return oldPrevious;
    }
};
IntStream.generate(fib).limit(10).forEach(System.out::println);
```

The code creates an instance of `IntSupplier`. This object has a *mutable* state: it tracks the previous Fibonacci element and the current Fibonacci element in two instance variables. The `getAsInt` method changes the state of the object when it's called so that it produces new values on each call. In comparison, our approach using `iterate` was purely *immutable*; you didn't modify existing state but were creating new tuples at each iteration. You'll learn in **chapter 7** that you should always prefer an *immutable approach* in order to process a stream in parallel and expect a correct result.

Note that because you're dealing with a stream of infinite size, you have to limit its size explicitly using the operation `limit`; otherwise, the terminal operation (in this case `forEach`) will compute forever. Similarly, you can't sort or reduce an infinite stream because all elements need to be processed, but this would take forever because the stream contains an infinite number of elements!

5.9. Overview

It's been a long but rewarding chapter! You can now process collections more effectively. Indeed, streams let you express sophisticated data pro-

ldition, streams can be parallelized

- The Streams API lets you express complex data processing queries. Common stream operations are summarized in **table 5.1**.

- You can filter and slice a stream using the `filter`, `distinct`, `takeWhile` (Java 9), `dropWhile` (Java 9), `skip`, and `limit` methods.
- The methods `takeWhile` and `dropWhile` are more efficient than a filter when you know that the source is sorted.
- You can extract or transform elements of a stream using the `map` and `flatMap` methods.
- You can find elements in a stream using the `findFirst` and `findAny` methods. You can match a given predicate in a stream using the `allMatch`, `noneMatch`, and `anyMatch` methods.
- These methods make use of short-circuiting: a computation stops as soon as a result is found; there's no need to process the whole stream.
- You can combine all elements of a stream iteratively to produce a result using the `reduce` method, for example, to calculate the sum or find the maximum of a stream.
- Some operations such as `filter` and `map` are stateless: they don't store any state. Some operations such as `reduce` store state to calculate a value. Some operations such as `sorted` and `distinct` also store state because they need to buffer all the elements of a stream before returning a new stream. Such operations are called *stateful operations*.
- There are three primitive specializations of streams: `IntStream`, `DoubleStream`, and `LongStream`. Their operations are also specialized accordingly.
- Streams can be created not only from a collection but also from values, arrays, files, and specific methods such as `iterate` and `generate`.
- An infinite stream has an infinite number of elements (for example all possible strings). This is possible because the elements of a stream are only produced *on demand*. You can get a finite stream from an infinite stream using methods such as `limit`.